WORKED SOLUTION

STEP I 1995, Question 4

Pure • De Moivre's Theorem

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s1-q4

The key idea

De Moivre's theorem turns $\sin 5\theta$ into the imaginary part of $(\cos\theta + \mathrm{i}\sin\theta)^5$, and the binomial theorem expands that power into products of $\cos\theta$ and $\sin\theta$. Every term of the imaginary part has an odd power of $\sin\theta$, so we can take out one factor and turn the rest into cosines with $\sin^2\theta = 1 - \cos^2\theta$. Then the angle $\frac{1}{5}\pi$ is special because $\sin 5\theta = 0$ there, which leaves a quadratic in $\cos^2\theta$.

Solution

The identity

Write $c = \cos\theta$ and $s = \sin\theta$. By de Moivre's theorem,

$$\cos 5\theta + \mathrm{i}\sin 5\theta = (c + \mathrm{i}s)^5.$$

By the binomial theorem, with coefficients $1, 5, 10, 10, 5, 1$,

$$(c + \mathrm{i}s)^5 = c^5 + 5c^4(\mathrm{i}s) + 10c^3(\mathrm{i}s)^2 + 10c^2(\mathrm{i}s)^3 + 5c(\mathrm{i}s)^4 + (\mathrm{i}s)^5.$$

Since $\mathrm{i}^2 = -1$, $\mathrm{i}^3 = -\mathrm{i}$, $\mathrm{i}^4 = 1$ and $\mathrm{i}^5 = \mathrm{i}$, the imaginary terms are those with odd powers of $\mathrm{i}s$:

$$\mathrm{Im}(c + \mathrm{i}s)^5 = 5c^4 s - 10c^2 s^3 + s^5.$$

Both c and s are real, so equating imaginary parts gives

$$\begin{aligned}
\sin 5\theta &= s\left(5c^4 - 10c^2 s^2 + s^4\right) \\
&= s\left(5c^4 - 10c^2(1 - c^2) + (1 - c^2)^2\right) \\
&= s\left(5c^4 - 10c^2 + 10c^4 + 1 - 2c^2 + c^4\right) \\
&= s\left(16c^4 - 12c^2 + 1\right),
\end{aligned}$$

using $s^2 = 1 - c^2$ in the second line. That is,

$$\sin 5\theta = \sin\theta\left(16\cos^4\theta - 12\cos^2\theta + 1\right),$$

as required.

Evaluating $\cos^2\frac{1}{5}\pi$

Put $\theta = \frac{1}{5}\pi$. Then $\sin 5\theta = \sin\pi = 0$, while $\sin\frac{1}{5}\pi \neq 0$ (as $0 < \frac{1}{5}\pi < \pi$). So we can divide the identity by $\sin\theta$, and $x = \cos^2\frac{1}{5}\pi$ satisfies

$$16x^2 - 12x + 1 = 0, \qquad \text{so} \qquad x = \frac{12 \pm \sqrt{144 - 64}}{32} = \frac{12 \pm 4\sqrt{5}}{32} = \frac{3 \pm \sqrt{5}}{8}.$$

? **Ask yourself** The quadratic has two roots. Which one is $\cos^2\frac{1}{5}\pi$?

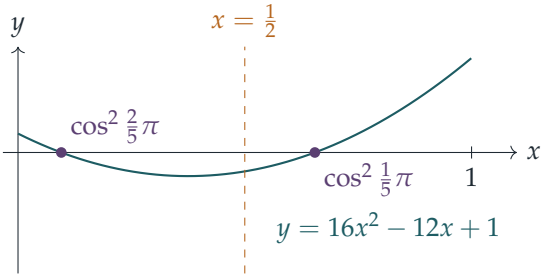
We need one more fact about $\cos\frac{1}{5}\pi$ to pick the root. Since $0 < \frac{1}{5}\pi < \frac{1}{4}\pi$ and cos is decreasing on $[0, \frac{1}{2}\pi]$, we have $\cos\frac{1}{5}\pi > \cos\frac{1}{4}\pi = \frac{1}{\sqrt{2}}$, so

$$\cos^2\tfrac{1}{5}\pi > \tfrac{1}{2}.$$

Now $\sqrt{5} < 3$ gives $\frac{3-\sqrt{5}}{8} < \frac{3}{8} < \frac{1}{2}$, while $\sqrt{5} > 1$ gives $\frac{3+\sqrt{5}}{8} > \frac{1}{2}$. Exactly one root is bigger than $\frac{1}{2}$, so

$$\cos^2\tfrac{1}{5}\pi = \frac{3+\sqrt{5}}{8}.$$

(The other root is not spurious: $\theta = \frac{2}{5}\pi$ also has $\sin 5\theta = 0$ and $\sin\theta \neq 0$, and $\cos^2\frac{2}{5}\pi = \frac{3-\sqrt{5}}{8}$.)



Deducing $\cos\frac{1}{5}\pi$

Since $0 < \frac{1}{5}\pi < \frac{1}{2}\pi$, $\cos\frac{1}{5}\pi > 0$, so we take the positive square root:

$$\cos\tfrac{1}{5}\pi = \sqrt{\frac{3+\sqrt{5}}{8}} = \sqrt{\frac{6+2\sqrt{5}}{16}} = \frac{\sqrt{6+2\sqrt{5}}}{4}.$$

Finally, $(1+\sqrt{5})^2 = 1+2\sqrt{5}+5 = 6+2\sqrt{5}$ and $1+\sqrt{5} > 0$, so $\sqrt{6+2\sqrt{5}} = 1+\sqrt{5}$ and

$$\cos\tfrac{1}{5}\pi = \tfrac{1}{4}\left(1+\sqrt{5}\right),$$

as required.

Checks

- At $\theta = \frac{1}{2}\pi$ the identity gives $\sin\frac{5}{2}\pi = 1\cdot(0-0+1) = 1$, which is right. Small θ: both sides are about 5θ, since the bracket tends to $16-12+1=5$.
- Numerically, $\cos 36^\circ \approx 0.809017$ and $\frac{1}{4}(1+\sqrt{5}) \approx 0.809017$; also $\cos^2 36^\circ \approx 0.654508 = \frac{3+\sqrt{5}}{8}$. We checked the identity at 10^4 random angles as well.
- $\frac{1}{4}(1+\sqrt{5})$ is half the golden ratio φ, the ratio of diagonal to side in a regular pentagon, as an angle of 36° suggests.

Common slips

- Forgetting the signs from powers of i: the s^3 term carries $\mathrm{i}^3 = -\mathrm{i}$, giving $-10c^2s^3$.
- Choosing the root of the quadratic without a reason. You need an inequality such as $\cos^2\frac{1}{5}\pi > \frac{1}{2}$ to rule out $\frac{3-\sqrt{5}}{8}$.
- Dividing by $\sin\theta$ without noting that $\sin\frac{1}{5}\pi \neq 0$.

Beyond STEP

The same method writes $\sin n\theta / \sin\theta$ and $\cos n\theta$ as polynomials in $\cos\theta$, the Chebyshev polynomials, which turn up throughout numerical analysis. The numbers $\mathrm{e}^{2\pi\mathrm{i}k/5}$ are the fifth roots of unity, and the fact that $\cos\frac{1}{5}\pi$ involves only square roots is why a regular pentagon can be constructed with ruler and compasses; Gauss's theorem on which regular polygons are constructible is a highlight of undergraduate Galois theory. For the geometry behind de Moivre's theorem, see Needham, *Visual Complex Analysis*.

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